Analytical Cume (February 23, 2001)
Theme: Separations, Capillary Electrophoresis

1. (42 points, 7 points for each) Capillary electrophoresis (CE), in general.

(a) What is capillary electrophoresis (CE) and how does it work?

(b) What are the important components of a CE apparatus?

(c) What kinds of substances may be separated by CE?

(d) What parameters of CE can be controlled to improve separations (to decrease plate height, H)?

(e) What parts of the Van Deemter equation are important in CE? Why?

(f) What advantages does CE have over slab gel electrophoresis?

2. (42 points, 14 points for each) A mixture contains 4 species, A, B, C, and D all of which are of spherical shape and whose radii increase directly proportionally to the molecular weight. "A" is a positive species (+1 charge) with a molecular weight of 240. B is a positive species with twice the charge (+2 charge), but with a molecular weight of 240. C is a neutral species with a molecular weight of 240. D is a negative species (-1 charge) with a molecular weight of 240.

For each question below, answer the following:
- What order of elution would you expect for the following CE configurations?
- At what end (anode= positive electrode, or cathode=negative electrode) would the molecules emerge?
- Explain your reasoning.

(a) (6 points) CE in a silica capillary with a typical aqueous buffer mobile phase.

(b) (6 points) CE in a silica capillary a cationic surfactant added to the mobile phase, in small amounts.

(c) (6 points) CE in a silica capillary with an anionic surfactant at a concentration higher than the critical micelle concentration is added to the mobile phase. What is the name of this separation method?

3. (6 points) What are three types of detectors for CE?

4. (10 points) How does electrophoretic separations on a microchip differ from traditional capillary electrophoresis? What are its advantages and disadvantages?